Partition Function and Thermodynamics of a Spin-$\frac{1}{2}$ System

Source: D. Tong, Statistical Physics, Problem Sheet 1, Question 3.

Consider a system consisting of N spin-$\frac{1}{2}$ particles, each of which can be in one of two quantum states, 'up' and 'down'. In a magnetic field B, the energy of a spin in the up/down state is $\pm\mu B/2$ where μ is the magnetic moment. Show that the partition function is

$$Z = 2^N \cosh^N\left(\frac{\beta\mu B}{2}\right). \tag{1.1}$$

Find the average energy E and entropy S. Check that your results for both quantities make sense at $T = 0$ and $T \to \infty$.

Compute the *magnetisation* of the system, defined by $M = N_\uparrow - N_\downarrow$, where $N_{\uparrow/\downarrow}$ are the number of up/down spins. The magnetic susceptibility is defined as $\chi \equiv$ \pd$M/$\pd$B$. Derive *Curie's Law*, which states that at high temperatures $\chi \sim 1/T$.

Solution. First, consider the partition function of a system with a single particle, z:

$$z = e^{-\beta\mu B/2} + e^{+\beta\mu B/2} = 2\cosh\left(\frac{\beta\mu B}{2}\right). \tag{1.2}$$

Assuming the particles are distinguishable (localised on a lattice), the partition function of a N particle system is, therefore,

$$Z = z^N = 2^N \cosh^N\left(\frac{\beta\mu B}{2}\right). \tag{1.3}$$

The average energy is given by

$$E = -\frac{\partial \ln Z}{\partial \beta} = -\frac{\mu B N}{2}\tanh\left(\frac{\beta\mu B}{2}\right), \tag{1.4}$$

and the entropy is

$$S = k_B(\ln Z + \beta E) = Nk_B\left[\ln 2 + \ln\cosh\left(\frac{\beta\mu B}{2}\right) - \frac{\beta\mu B}{2}\tanh\left(\frac{\beta\mu B}{2}\right)\right]. \tag{1.5}$$

At $T = 0$, $\beta \to \infty$: $\ln\cosh(\beta\mu B/2) \approx \beta\mu B/2$ and $\tanh \to 1$, so the last two terms cancel and $S \to 0$, consistent with a unique ground state. For $T \to \infty$, $\beta \to 0^+$: $\ln\cosh \to 0$ and $\tanh \to 0$, giving

$$S \to Nk_B \ln 2, \tag{1.6}$$

the maximum entropy corresponding to all 2^N microstates being equally probable.

To compute the magnetisation, let us go back to consider the single particle system. The average magnetisation of such a system is given by

$$\langle M \rangle = \frac{1}{z}(e^{-\beta\mu B/2} - e^{\beta\mu B/2}) = -\tanh\left(\frac{\beta\mu B}{2}\right). \tag{1.7}$$

The magnetisation of a N-particle system is simply

$$M = -N\tanh\left(\frac{\beta\mu B}{2}\right). \tag{1.8}$$

For large temperature, $\beta \to 0^+$, which gives

$$\chi \approx -\frac{N\beta\mu}{2} \sim \frac{1}{T}. \tag{1.9}$$